

20 April 2026

#chapter 10

As per last class,

Class diagram

Activity diagram

Sequence diagram

Statechart diagram

UML Diagram

Use Case Scenario

Use Case Diagram

Method Overloading

↳ same name

different parameters

ECB Sequence Diagram

↑ ECB কয়?!

Types of classes // based on structure & behavior

Entity class

Interface class

Abstract class Not needed

Control class

Entity → Data store, initiate

Interface → Interaction with framework/system (UI)

Abstract → Direct object create করা যায় না

Control → Logic control/implementation/work flow

ECB = Entity Control Boundary

Interface ~ Boundary

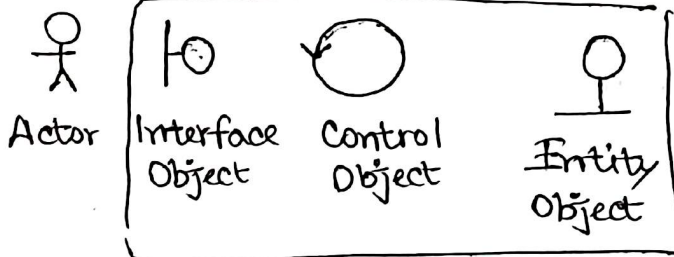
Key Components of ECB Sequence Diagram

① Actor : initiates interactions.

② Boundary Object (Interface) : interacts with actor

③ Control Object

④ Entity Object : represents persistent data

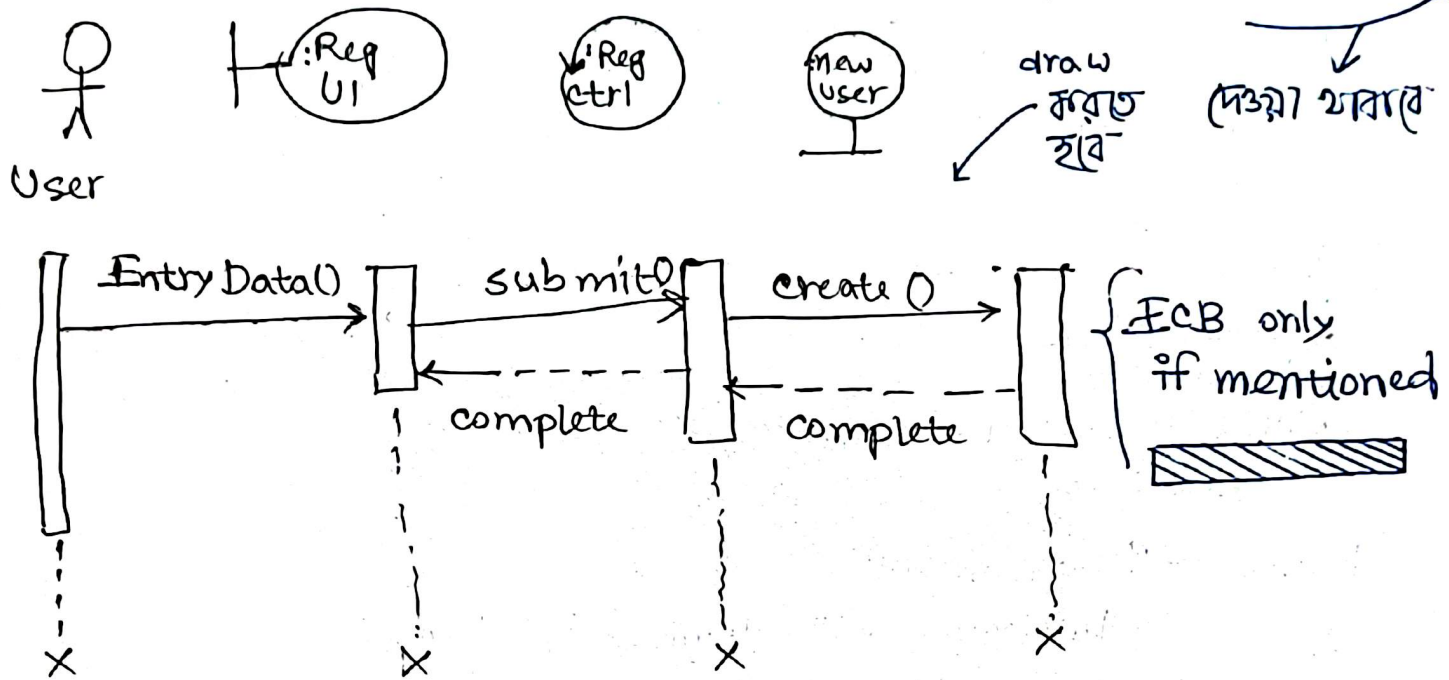


At least 1 to category there
1 to করে থাকবে
= 3 টি

Sequence Diagram Example User Registration

Shayan

Actor (User): enters data into boundary (Registration UI)
Boundary sends a submit message to Control (Registration Controller)
Control creates a new Entity (User) instance
Control sends confirmation to Boundary to update the display.



Chapter II

Designing Output

TPS = Transaction Processing System
DPS = Data Processing System

DSS = Decision Support Systems
MIS = Management Information Systems

Categories of Information

- Strategic: E.g. Trends in Revenue
 - Management: E.g. Inventory Information
 - Operation: E.g. TPS/DPS
- managerial

Purposes or Objective of Business Reports

- ① Transmitting Information
- ② Interpretation and Explanation of event
- ③ Making decisions

- ④ Communication with External Stakeholders
- ⑤ Development of information base
- ⑥ Developing labor-management relationship

Someone
to laugh with
(comedian)
good
humour

Someone
to laugh at
(clown)
Bad
humour

* Same information
different presentation

- ⑦ Controlling
- ⑧ Recommended Actions

Output Design Objectives

- ① Usefulness & Acceptance
- ② Serves specific user or organizational purpose
- ③ Meaning to the user
- ④ Deliver appropriate quantity of output
- ⑤ Ensure the output is where it is needed
- ⑥ Provide output on time
- ⑦ Choosing the right output method

Output Design Advantages/Disadvantages

Self study